Comparative Analysis of Machine Learning Techniques for Mental Health Prediction

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***Abstract*—The prevalence of mental health problems has prompted investigations into the use of machine learning to tackle the issues. Mental health is a crucial component of an individual's overall well-being and can be detected and treated early on, significantly improving the quality of life for those affected. This study examines the use of machine learning algorithms to predict mental health disorders using a dataset of self-reported information. Four commonly used machine learning models K-nearest neighbor classifier, logistic regression, random forest and decision tree are compared in terms of their performance. The objective of this study is to compare the performance of these machine learning algorithms on a self-reported mental health dataset and identify the most suitable model for predicting mental health. The challenges faced by the system include the limited size and quality of the dataset, the need for ethical considerations in handling sensitive mental health information, and potential biases in the data .The results of the experiments identify the most suitable model for predicting mental health.**

***KeyWords*—Logistic regression, K-Nearest Neighbors, Decision tree , Mental Health, and Random forest**

I. INTRODUCTION

Millions of individuals around the world are affected by the serious problem of mental health. Nearly one out of every four persons will suffer from a mental or neurodegenerative ailment at a certain stage of their lives, according to the World Health Organization (WHO). The goal of the fast-expanding field of research known as "mental health prediction using machine learning "is to diagnose and treat mental health illnesses. There have been several research studies published on this topic in recent years, which have used various machine learning techniques, such as support vector machines, artificial neural networks, and decision trees to predict mental health outcomes based on a variety of data sources. Early identification and prediction of mental health disorders can help in their management and treatment. Algorithms for machine learning can increase the accuracy and efficiency of mental health forecasting. Mental health has grown into a major concern in the modern society, with increasing cases of depression, anxiety, and other mental illnesses. The data sources used in these studies include electronic health records, social media data, wearable devices, and self-reported surveys. The features extracted from these data sources include demographic in-formation, clinical symptoms, medication use, social support, physical activity, sleep patterns, and language use. The goal of these studies is to develop accurate and reliable prediction models that can help healthcare professionals to identify and intervene with patients who are vulnerable to issues with mental health.

There are various difficulties with utilising machine learning to predict mental health, including data quality, privacy concerns, and model interpretability. It is also important to ensure that the models are generalizable and applicable to diverse populations. Despite these challenges, machine learning-based mental health prediction has the potential to revolutionize the field by enabling early and accurate diagnosis of mental health disorders, enhancing the efficacy of the therapy, and reducing healthcare costs. However, it is crucial to develop and evaluate these models responsibly and ethically while ensuring their accessibility and inclusivity. Early identification and treatment of mental health disorders can improve individuals' quality of life. Machine learning techniques such as classification algorithms can be used to predict mental health conditions based on various factors. This paper examines the use of k-nearest neighbor classifier, logistic regression, random forest, and decision tree models for predicting mental health disorders.

II. LITERATURESURVEY

Numerous research has been looked into using machine learning algorithms for predicting psychological results.

Jason Teoet.al,[1]examined the prospect of machine learning in tackling psychological disorders but also showed the hurdles and restrictions experienced by researchers in this area. Machine learning algorithms and feature selection can have a big impact on classification performance, and newer methods that use different sensor modalities show promise. Machine learning models might not perform consistently across various challenges, and more externally, validated data is required. For obtaining the best results, preprocessing tasks like data cleaning and parameter adjustment are also crucial. To address these problems, researchers must thoroughly explore and evaluate data with various machine learning algorithms and control restrictions to get outcomes that can enhance medical care.

Deep learning techniques are used in a different study [2] which focused on the use of social media for early depression identification. The authors talked about earlier research that examined depression and other mental health conditions using information from social media. These studies show how social media has the potential to serve as a source of data for the early identification of mental health problems. The authors discussed how to detect depression using deep learning methods like convolutional Neural networks (CNNs) and long short-term memory (LSTM)networks. They emphasized the benefits of deep learning over conventional machine learning methods for deciphering complicated data, such as that from social media.

In a different study [3], Machine learning prediction of depression and anxiety from electronic health records and a novel machine learning approach with artificial intelligence was proposed. The study looked at how to predict sadness and stress using electronic health records using machine learning approaches. The authors reviewed earlier research that employed machine learning methods to predict mental wellness, including depression and anxiety. They highlighted the potential of machine learning algorithms in identifying predictive factors for mental health disorders and providing personalized treatment recommendations. The authors discussed the limitations of their study, including the use of a single HER database and the lack of external validation. They also highlight future directions for research, such as developing more robust and interpretable machine learning models and integrating other sources of data , such as genomics and social media.

Additionally, in various research, machine learning methods have been used to identify mental health issues. A study by KiranSaqibet.al.,[4] discussed the prevalence, risk factors, and consequences of PPD (Paranoid personality disorder). They also highlighted the importance of early identification and intervention for PPD to improve maternal and child health outcomes. The authors reviewed studies that have used machine learning methods for PPD prediction. They discussed the methods used in these studies, the features selected for prediction, and the performance of the models developed. The authors highlighted the limitations of existing studies, such as the lack of diverse datasets and the limited use of advanced machine-learning techniques.

Johannes Lieslehto et.al.,[5] discussed the potential of machine learning in addressing psychological disorders but also emphasized the difficulties and restrictions that researchers confront in this area. Performance in classification may be considerably impacted by the features and methods employed in machine learning, and newer approaches that integrate various sensor modalities show promise. However ,there is a need for more validated evidence from external sources, and machine learning models may not perform consistently across all problems. For attaining the best results, preprocessing operations like parameter adjustment and data cleaning are also crucial. Researchers should closely review, evaluate data using a variety of machine learning methods, and handle constraints in order to provide outcomes that might enhance clinical practise and decision-making.

Danxia Liu, et al., [6] explored the findings of studies that have demonstrated the effectiveness of machine learning approaches for leveraging text information gathered via social media to identify anxiety. Nevertheless, further study is required to address concerns including sample size, optimisation of prediction algorithms and characteristics, generalizability, privacy, and ethics. These methods can supplement conventional ways for diagnosing depression. Overall, the paragraph suggests that ML approaches have the potential to be a valuable tool for depression detection and diagnosis, but more work is required to fully realize their benefits.

Aziliz Le Glaz et.al.,[7] aimed to identify the different applications of ML and NLP in the field of mental health, the types of data used, and the performance of the methods used in various studies. The authors found that the use of ML and NLP in mental health has increased significantly over the years and has been used in a variety of contexts, including diagnosis, prognosis, and treatment. The review also highlighted the challenges and limitations of using ML and NLP in mental health, such as the lack of standardized data, privacy concerns, and the need for more extensive validation of the results. Overall, the authors concluded that ML and NLP have the potential to significantly impact mental health research and clinical practice, but more work is needed to overcome the challenges and limitations.

Safwan W shah et.al.,[8] analysed information from the National Epidemiologic Survey on Alcohol and Associated Disorders (NESARC) to train and test the machine learning model. The model combines demographic information, information on the traumatic event, and information on mental health symptoms to predict the risk of developing PTSD.The results of the study showed that the machine learning model was able to predict the risk of PTSD with high accuracy.The study demonstrated machine learning's ability approaches to improve the prediction and prevention of mental health disorders.

Ashley E et.al.,[9] examined machine learning methods for detecting mental health problems in adolescents. The study uses data from the Swedish Twin Registry, comprising over 30,000 individuals, to train and test several machine learning models. The models use a combination of demographic, familial, and behavioral factors to forecast the likelihood of psychological disorders in adolescence. The results of the study show that machine learning algorithms can predict the likelihood of psychological disorders with high accuracy, outperforming traditional statistical models. The study highlights the potential of machine learning techniques for identifying individuals at risk of psychological disorders and could aid in the early identification and prevention of mental health disorders in adolescents.

Chang Su et.al.,[10] presented a thorough analysis of the state of deep learning techniques in mental health outcome research. The authors conduct as coping review of the literature and identify 37 relevant studies that utilize deep learning methods in predicting and diagnosing mental health outcomes. The authors summarize the findings of these studies and highlight the advantages and drawbacks of the deep learning methods employed in each domain.

In another study, Senerath Mudalige et.al.,[11] delve into how bioinformatics uses machine learning methods, This has made a substantial contribution to the medical sector in terms of quick, accurate, precise, and economical computerized decision-making. Machine learning is used in a variety of applications, including biomedical event extraction, medication repurposing, illness diagnosis and prediction, medical imaging, and more. The study underlines how decision-making in the medical field has used artificial intelligence as a crucial resource in a variety of ways,and its recent applications during the COVID-19 pandemic have highlighted its importance. The research comes to the conclusion that artificial intelligence and machine learning have been extremely important in the implementation of computerized decision-making tools for the health sector.

In another article, Geng-Shen Fu et al., [12] reviewed various ML techniques, including supervised, unsupervised, and reinforcement learning, that have been utilized for medical image analysis. The authors first discuss the prospects and limitations of ML in medical imaging. Convolutional neural networks(CNNs), a form of deep learning technique ,and their applications in a variety of medical specialties, including radiology, cardiology, and ophthalmology, are also discussed. The authors additionally stress the necessity of data preprocessing, feature extraction, and model interpretation in constructing efficient ML models for medical imaging analysis. The paper highlights the promise of ML techniques in improving diagnosis, prognosis, and treatment planning in clinical practice and offers a thorough has sessment of the state-of-the-art of current ML applications in medical imaging.

Stevie et.al.,[13] discussed the importance of in corporating human-centered approaches in machine learning for forecasting mental health outcomes from social media data. The authors argue that while machine learning algorithms have shown promise in detecting mental health concerns, they often overlook the social and contextual factors that shape mental health. The paper presents three case studies that demonstrate how in corporating human-centered approaches, such as incorporating expert knowledge and in corporating social context, can improve the accuracy and reliability of machine learning models for predicting mental health outcomes. The authors arrive at the opinion that human-centered methods are essential for developing efficient and ethical algorithms for machine learning in the field of mental health.

Anja Thieme et.al.,[14] provideed a exploratory review of the Human-Computer Interaction (HCI) study which can aid in the creation of powerful and practical machine learning (ML) frameworks. The authors investigate how the HCI community has addressed various challenges associated with developing ML systems and how they have evaluated their effectiveness. The paper highlights key findings, including the importance of user-centered design, and transparency, and explains the ability of ML systems. It also identifies gaps in the current literature, such as the need for more studies on the social impact of ML and the need for better evaluation methods for assessing the effectiveness of ML systems. The authors conclude by offering recommendations for future research to address these gaps and provide guidelines for developing effective and implementable ML systems that are sensitive to human needs and values.

Gyeongcheol Cho et.al.,[15] provides a summary of several machine learning techniques for detecting mental disorder.
The authors discussed the need for such algorithms, citing the increasing prevalence of mental health disorders and the inadequacy of traditional diagnostic methods. They then describe the various types of machine-learning algorithms, including decision trees, support vector machines, and deep learning, and explain how they can be applied to diagnose mental illness. The authors also review researches that have utilized machine learning algorithms for mental health diagnosis, highlighting their successes and limitations. They conclude by suggesting areas for future research and emphasizing the importance of interdisciplinary collaboration between mental health professionals and machine learning experts.

Anna Markella et.al.,[16] analysed the challenges associated with data collection, storage, and sharing in healthcare, highlighting the need for standardized and interoperable data formats. They also addressed the ethical and legal concerns regarding data privacy and security. The paper then discusses the challenges in developing and validating ML models in healthcare, such as limited sample sizes, bias, and interpretability. The authors also discuss potential future directions for ML in healthcare, including the development of personalized medicine and the integration of ML into clinical decision-making processes. They highlight the importance of interdisciplinary collaborations between ML researchers, healthcare providers, and patients to ensure the successful implementation and adoption of ML techniques in healthcare.

Sumathi et.al., [17] conducted a study to predict mental health problems among children using machine learning techniques. The study involved collecting data from 200 school-going children in India, using the Strengths and Difficulties Questionnaire (SDQ) as the basis for assessing mental health problems. The authors then used various machine learning approaches, including Naive Bayes, Decision Tree and k-Nearest Neighbor algorithms, to forecast children's mental health issues based on their SDQ scores. The outcomes demonstrated that the Decision Tree algorithm performed better than the competing algorithms, with an accuracy of 85%. The report's findings suggested that machine learning methods may be used to predict children's mental health issues and that early treatment can help keep these issues from getting worse.

III. METHODOLOGY

The process of finding accuracy and predicting results from data includes several steps such as cleaning, data collection, finding the co-variance matrix, encoding, fitting, scaling, tuning, evaluating models and discovering knowledge from data as shown in the Fig 1below.. The dataset being considered has 1260 rows and 27 columns. Data cleaning involves identifying and addressing any incorrect, incomplete, extra or missing data by updating, modifying or removing it as needed.

In the data cleaning process, it was discovered that three columns contain missing data. In Data Frames and Numpy arrays, cells with no value are represented by the special value NaN. The subsequent phase is data encoding, which is used when a categorical feature is represented as an integer. Correct order must be maintained throughout the encrypting. Each label undergoes an integer value conversion throughout the label encrypting procedure. After that, the covariance matrix is calculated. It is a fundamental matrix used in data science and machine learning.

The covariance matrix provides information about the correlation between features. The ranges of the values are contained in the variance-covariance matrix's diagonal elements, while the other matrix positions contain the covariances between pairs of variables. The means vector contains the means of each variable. The individual properties of the data are scaled by placing them within a predefined range to manage drastically changing values, units or magnitudes during data pre-processing. The dataset is then split into training and testing datasets. In machine learning, feature selection is crucial since it directs the usage of variables to what will be most beneficial and efficient for a particular system using machine learning. The next stage is tuning, which entails choosing the right hyper-parameters to enhance model performance while avoiding overfitting or excessive variation, It may be compared to the "knobs" or "dials" of a machine-learning algorithm. After that, the models are assessed using a variety of machine learning methods, including K-nearest neighbour classifier, logistic regression, random forest classifier and decision tree classifier.

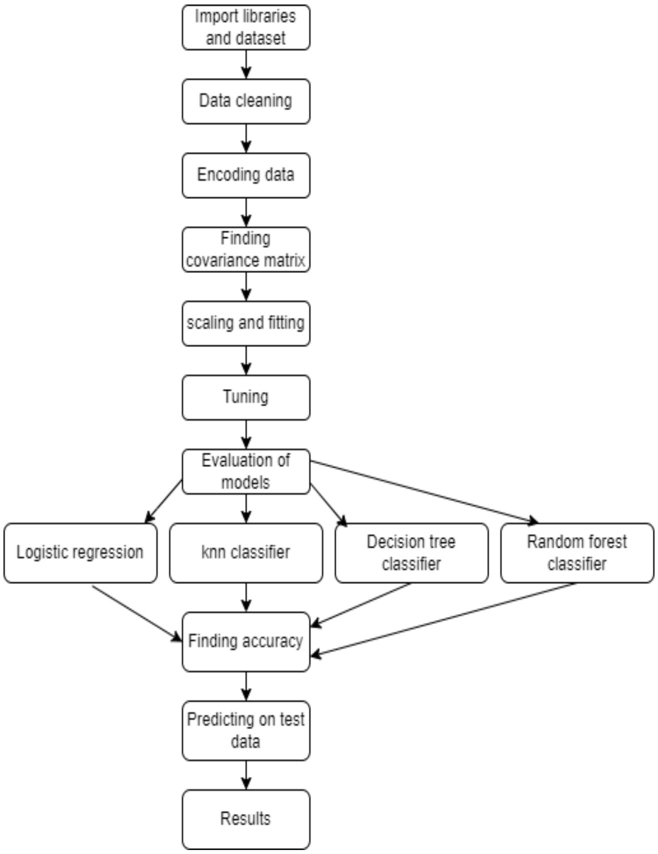

Fig1. Architecture

A) Logistic Regression: This is a statistical method used in machine learning and data science to solve classification problems. It is a form of regression analysis that is applied when the dependent variable is binary or categorical, meaning it can only have two possible outcomes. The goal of logistic regression is to determine the relationship between independent variables and the probability of a particular outcome. The result of a logistic regression model is a predicted probability of an event occurring, which always falls between 0 and 1.

B) K-Nearest Neighbors Classifier: A popular machine learning method for classification and regression issues is K-Nearest Neighbors (KNN). This approach is non-parametric, i.e., it does not assume anything on how the data are distributed. In KNN classification, the aim is to classify unknown data points by comparing them to the K nearest data points in the training set. The K nearest data points are selected based on their distance from the unknown data point, which can be calculated using various metrics such as Euclidean or Manhattan distance.

c) Decision Tree Classifier: This is a popular machine learning technique used for both regression and classification problems. It is a tree-based model that makes decisions based on a hierarchy of if-then rules. The decision tree method recursively divides the data into smaller groups based on the most useful features for the classification problem, resulting in a tree-like model. However, decision tree classifiers can be biased towards features with more levels or those that appear earlier in the tree. This can be addressed by using techniques such as random forests.

D) Random Forest Classifier: This is an ensemble learning method that combines multiple decision trees to improve the robustness and accuracy of the classification model. This popular machine learning technique can be used to solve both classification and regression problems. In a random forest classifier, several decision trees are trained on different subsets of the training data, using a random subset of features at each split. This randomness reduces the correlation between the trees, improving the model's performance and reducing overfitting. The final prediction of the random forest classifier is determined by taking the majority vote of the individual tree predictions.

```
Randomforest:
Rand. Best Score:  0.8305206349206349
Rand. Best Params:  {'criterion': 'entropy'
[0.835, 0.831, 0.831, 0.831, 0.831, 0.831,
Accuracy: 0.8121693121693122
0    191
1    187
```

Fig2. RandomForest Accuracy

```
Decision tree
Rand. Best Score:  0.8305206349206349
Rand. Best Params:  {'criterion': 'ent
[0.828, 0.83, 0.831, 0.831, 0.831, 0.8
Logistic regression:
Accuracy: 0.8068783068783069
```

Fig3. Decisiontree Accuracy

```
Logistic regression:
Accuracy: 0.7962962962962963
0    191
1    187
Name: treatment, dtype: int64
Percentage of ones: 0.4947089947089947
Percentage of zeros: 0.5052910052910053
```

Fig4. Logistic Regression Accuracy

```
Rand. Best Score:  0.8201841269841269
Rand. Best Params:  {'weights': 'uniform'
[0.823, 0.819, 0.819, 0.82, 0.815, 0.818,
knn:
Accuracy: 0.798941798941799
0    191
1    187
```

Fig5. KNN Accuracy

IV. RESULTS

All four models were successful in predicting mental health issues. The ROC curves for Random Forest, Logistic regression, KNN and Decsion tree are shown in Fig 6-9. Similarly table 1 shows the accuracy obtained by the four models analyzed in this study. The logistic regression model had an accuracy of 79%, with 0.76 precision, 0.77 recall, and 0.71 f1-score. The k-nearest neighbor classifier achieved an accuracy of 80%, with 0.74 precision, 0.73 recall, and 0.69 f1-score. The decision tree model had an accuracy of 79%, with 0.74 precision, 0.63 recall and 0.65 f1-score. The random forest model had the highest accuracy at 81%, with precision of 0.75, recall of 0.74, and an F1 score of 0.76. The accuracy scores of the classifiers were relatively close to each other, with a maximum difference of only 2 percentage points between the best and worst-performing classifiers. The decision tree and random forest classifiers had slightly higher accuracy scores than the logistic regression and k-nearest neighbor classifiers, suggesting that these models may be better suited to this particular dataset.

TABLE I ACCURACY

Classifier	Accuracy
logisticregression	79%
k-nearestneighbor	79%
decisiontree	80%
randomforest	81%

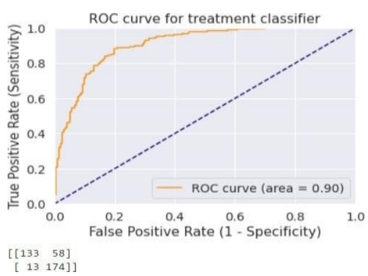

Fig6. Random Forest ROC curve

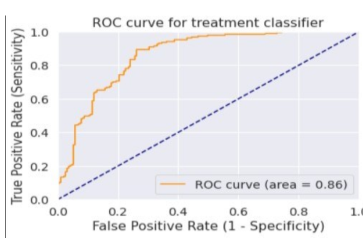

Fig7. Logistic Regression ROC curve

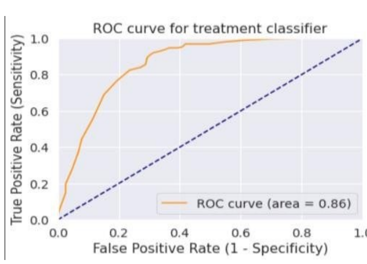

Fig8 . KNN ROC curve

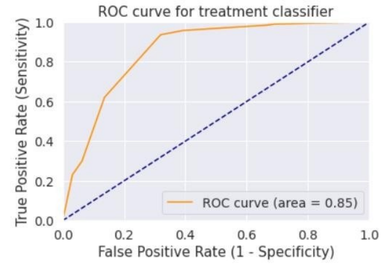

Fig9. Decision tree ROC curve

V. FUTURE SCOPE

In the future, machine learning can be used in conjunction with other technologies such as wearable devices, mobile apps, and virtual reality to provide a more comprehensive and personalized approach to mental health treatment. This can enable early intervention for mental health disorders, potentially preventing or reducing the severity of symptoms. The use of multimodal data sources such as speech, text, and physiological data can improve the accuracy of mental health prediction models. Future research can focus on developing models that predict the effectiveness of different treatment options for individual patients, allowing for personalized treatment plans. The interpretability of Moreover, models based on machine learning may be enhanced, making it easier for healthcare providers to understand how the model arrived at its predictions.

VI. CONCLUSION

In this study, survey data is used to predict mental health illnesses using algorithms that employ machine learning. A csv dataset was investigated. The performance of four popular machine learning models: logistic regression, k-nearest neighbor classifier, decision tree, and random forest were compared. The results showed that all four models performed well in predicting mental health disorders, with the random forest model achieving the highest accuracy. According to these findings, algorithms that use machine learning can increase the precision and effectiveness of mental health diagnosis, but further research is needed to fully understand their potential in this field.

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